Feedback for “ICS 604: Applied Data Science – Final Project Proposal”

The research question is clearly stated, focused, and well defined. The research question was “Do NBA prediction markets exhibit systematic pricing bias correlated with media attention disparities?”. The hypothesis was “teams receiving disproportionate media coverage are systematically overpriced in Kalshi markets relative to their actual win probabilities”. The steps to test the hypothesis are well-structured, detailing how media attention will be quantified, how pricing errors will be calculated, and which statistical methods will be used to evaluate the effect.

The project scope is reasonable given the timeframe, as the analysis is limited to NBA games from the 2025 season with corresponding Kalshi markets, totaling approximately 1,000–3,000 games. The necessary data from Kalshi, Basketball-Reference, and GDELT is available. While processing large datasets and filtering and aligning GDELT news coverage may require extra effort, the scope is manageable.

The proposed methods are appropriate for addressing the research question. Feature engineering will create a composite media disparity score to quantify media attention differences. Linear regression will model pricing error as a function of the media disparity score while controlling for confounding factors to determine whether differences in media coverage systematically predict over or underpricing in NBA prediction markets. Bootstrap resampling will provide confidence intervals and test statistical significance of the media bias effect. Time series analysis will examine how market prices evolve and whether these movements correlate with media coverage, offering insight into media-driven market dynamics.

The datasets are appropriate and sufficient for the proposed analysis. The Kalshi dataset provides market-implied probabilities and actual outcomes for the NBA games from January-October 2025. Basketball-Reference offers comprehensive team statistics, and GDELT provides media coverage and sentiment data needed to quantify media disparity.

One suggestion to mitigate the potential risk of running out of time is to prioritize the linear regression and bootstrap analyses, and conduct the time series analysis only if time permits. Given the large datasets involved and the potential time required to preprocess and filter GDELT media coverage data, focusing on the core analyses will help ensure the project remains manageable within the timeframe. Another suggestion is to consider setting a minimum threshold for GDELT media coverage. Teams with very low media attention can still provide meaningful insight into how markets behave under minimal coverage, but extremely sparse mentions may produce noisy or unstable media disparity scores. Filtering only these extreme low-coverage cases can improve robustness while preserving the ability to analyze informative low-coverage scenarios.

A strength of the proposal is its clear, step-by-step plan for testing the hypothesis. It outlines how media attention will be quantified, how pricing errors will be calculated, and which statistical methods will be used, making the approach logical and replicable. Another strength is the strong motivation and real-world relevance, exploring whether media sentiment can meaningfully influence market pricing in ways that diverge from actual outcome probabilities.